Estimated Position of the Moving Axes of a Mechanical Arm from an Unknown State

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Abstract. This work presents an improved methodology for monocular robot pose estimation by enhancing a holistic pose estimation framework through the integration of depth-aware feature extraction and self-supervised learning. The proposed approach incorporates a depth estimation network to extract complementary geometric information from RGB images, generating enriched feature representations through depth maps and heatmaps. These features are concatenated and compressed before being used in the original pose estimation architecture, preserving the network structure while improving representation quality. The training process consists of two stages. First, the model is trained with synthetic data to learn geometric relationships between robot joints, camera parameters, and 3D keypoints. Then, the model is refined using self-supervised learning with real data, guided by kinematic and segmentation consistency constraints instead of ground-truth labels. The self-supervised strategy uses forward kinematics reconstruction and differentiable rendering to maintain geometric consistency between predicted robot pose and image observations. Experimental results show that the proposed method achieves competitive performance in robot pose estimation tasks, slightly outperforming previous methods while maintaining stable training and strong generalization to real-world scenarios. These results indicate that combining synthetic pretraining, depth-enhanced features, and self-supervised adaptation is effective for improving monocular robot pose estimation.

Keywords: Robot Pose Estimation, Monocular Depth Estimation, Computer Vision, Deep Learning, Convolutional Neural Networks, Keypoint Detection Depth-Aware Feature Fusion.

1 Introduction

In recent years, Artificial Intelligence (AI) has become an essential tool in daily activities across society and industry (Pratti, 2025). Among its research areas, computer vision has stood out as a key driver of technological advances, particularly in healthcare, automation, and robotics (Barbosa, 2023).

In robotics, several recent studies focus on estimating robotic manipulator pose using RGB images combined with joint state information. Methods such as CtRNet-X and

RoboPose have demonstrated high accuracy in pose estimation tasks (Lu, 2024). However, the need for prior knowledge of joint angles limits these approaches, reducing their applicability in scenarios where such data is unavailable or unreliable (Lu, 2023). As a result, there is an increasing demand for models capable of accurately estimating pose using only visual information (Ban, 2024).

In this context, this research investigates the use of Convolutional Neural Networks to estimate the pose of a 7-DOF articulated robotic manipulator (Panda model). Based on the estimated pose, the proposed method also performs robot segmentation in the input image using forward kinematics and camera intrinsic parameters.

2 Methodology

2.1 Intrinsic and Extrinsic Matrices

Among the various techniques and models employed in computer vision, the projection of 3D points onto image coordinates plays a central role in keypoint estimation using a single RGB image as input.

Within the mathematical models available in the literature, the pinhole camera model stands out due to its simplicity and effectiveness. In this model, a point in the world coordinate system

$$W = (X, Y, Z)^T \tag{1}$$

is mapped onto the image plane as

$$M = \left(\frac{fX}{Z}, \frac{fY}{Z}, Z\right)^T \tag{2}$$

according to the following projection relationship:

$$M = \mathbf{K}\,[\mathbf{R} \mid \mathbf{t}]\,W \tag{3}$$

where K represents the intrinsic camera matrix, which encapsulates the internal parameters of the camera, such as focal lengths and the coordinates of the principal point:

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \tag{4}$$

In this matrix:

f_x: denotes the focal length along the x-axis;

f_y: denotes the focal length along the y-axis;

c_x: represents the x-coordinate of the principal point;

c_y: represents the y-coordinate of the principal point.

The matrices **R** and **t** compose the extrinsic parameters, which define the rotation and translation of the camera with respect to the world coordinate system, respectively

(Waleed, 2024). To ensure a geometrically consistent projection, the rotation matrix satisfies $\mathbf{R} \in SO(3)$;

2.2 Depth Anything

Obtaining depth information directly from RGB-D cameras is not always viable due to hardware limitations or high equipment costs, which makes depth data unavailable in many real-world scenarios. To address this issue, monocular depth estimation methods have been proposed, with Depth Anything V2.0 standing out as a state-of-the-art approach for estimating depth from single RGB images.

Depth Anything V2.0 is trained on more than 60 million images using a Teacher–Student learning framework. The model uses DINOv2 as a feature extractor, whose outputs are processed by the DPTHead module to refine features and generate the final depth map through convolution and interpolation operations (Yang, 2024).

In this framework, a teacher model based on DINOv2-Giant is first trained using labeled synthetic data. This model then generates pseudo-labels for real-world images, which are used to train multiple student models based on DINOv2-Small, Base, and Large. This strategy enables knowledge transfer from synthetic data to real-world scenarios, improving depth estimation performance (Yang, 2024).

2.3 HRPE Model

Currently, the Holistic Robot Pose Estimation (HRPE) model (Ban, 2024) has the best metrics in the literature when we talk about the Deep Robot-to-Camera Extrinsics for Articulated Manipulators - DREAM dataset (Lee, 2020), whose main objective is to provide a robust and diverse dataset for estimating the pose of a robotic arm from RGB images.

Originally, the model consists of 4 distinct neural networks: DepthNet, JointNet, RotationNet, and KeypointNet, as shown in Figure 1.

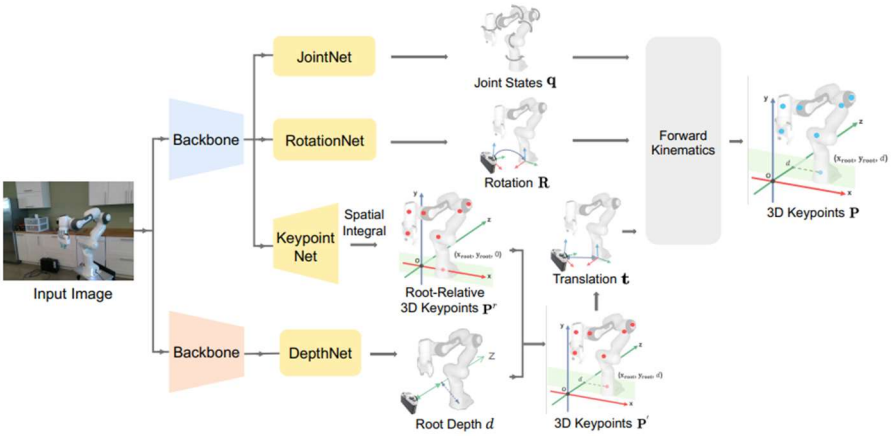


Fig. 1. General structure of the Holistic Robot Pose Estimation (HERP) model (Ban, 2024)

Starting from an RGB I image, we use each module of this network set to perform specific tasks, which are:

- DepthNet: Estimate the absolute depth d of the root;
- JointNet: Retrieve joint status parameters **q**;
- RotationNet: Predicts the camera rotation matrix **R** relative to the robot;
- KeypointNet: Calculate 3D positions relative to the root of the key points $\mathbf{P_r} \in R^{Nx}$, where N corresponds to the total number of joints present in the robotic arm.

Next, we combine the results obtained from DepthNet and KeypointNet, and obtain the absolute positions of the keypoints $\mathbf{P'} \in R^{Nx3}$ that are used to determine the translation t. Then, the joint parameters q, along with R and t, are applied in the direct kinematic process to obtain the final 3D estimates of the keypoints $\mathbf{P} \in R^{Nx}$, (Ban, 2024).

The JointNet, RotationNet, and KeypointNet networks share the same Resnet-50 feature extractor, while the DepthNet network uses the more robust Resnet-32 extractor.

DepthNet. The DepthNet network is a single-layer convolutional network that seeks to estimate the distance between the base of the arm and the camera. However, the estimation of the depth of monocular objects suffers from significant ambiguities, since the absolute depth value and the camera's intrinsic values are deeply intertwined. Therefore, the use of DepthNet is divided into two parts. Initially, using the camera's intrinsic data, we calculate a normalized depth value (Ban, 2024).

$$\mathbf{d} = \sqrt{\frac{f_x \cdot f_y \cdot A_{real}}{A_{bbox}}} \tag{5}$$

In this equation, A_{real} represents the robot area projected onto the camera x–y plane in mm², while A_{bbox} corresponds to the area of the robot bounding box in pixels².

Thus, we use the convolutional network only to estimate a correction factor **λ** from the features $\mathbf{f_1}$, which, once multiplied by **d**, can provide the robot's absolute root depth.

$$pred_{depth} = \boldsymbol{\lambda} \cdot \mathbf{d} \tag{6}$$

JointNet. Through several MLP blocks, we use the extracted features $\mathbf{f_2}$ to estimate the angle in radians of each joint of the arm (Ban, 2024).

RotationNet. Similarly, in RotationNet we aim to estimate the camera-to-robot rotation matrix through MLP blocks using the same features f_2 (Ban, 2024).

KeypointNet. The objective of this convolutional network is to estimate the location of the keypoints relative to the root point in an R^{Nx3} space, where N corresponds to the number of robot joints, using transposed convolutional networks.

2.4 New Model

The methodology adopted to improve the HRPE model (Ban, 2024) consists of incorporating the Depth Anything network (Yang, 2024) in order to explore depth information and generate heatmaps from the images, enriching the feature representation.

Thus, we concatenate and compress the new features and use them to train the model while maintaining the network architecture, as shown in Figure 3.

In addition, we seek to optimize the model's performance by modifying its hyperparameters, training approach, and reducing the amount of data during training with synthetic values.

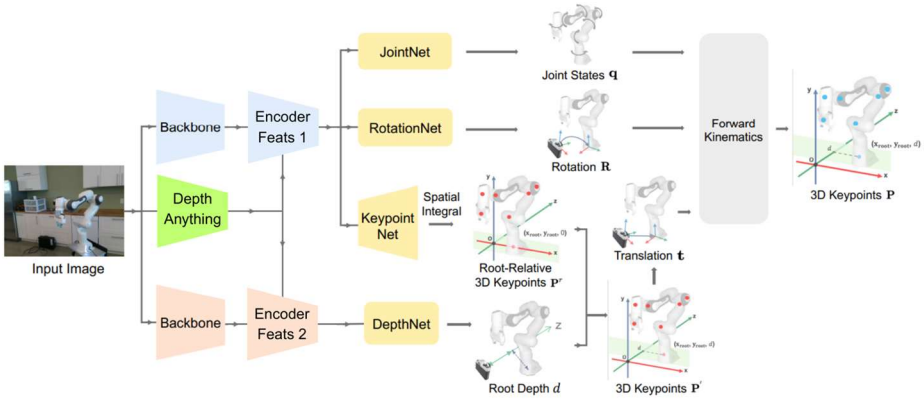


Fig. 2. Overview of the proposed methodology for enhancing the HRPE model (Ban, 2024) through the integration of the Depth Anything network (Yang, 2024). The pipeline illustrates the extraction of depth information and the generation of heatmaps from input RGB images, which are incorporated to enrich the feature representation. The newly extracted depth-aware features are concatenated with the original features and subsequently compressed before being fed into the HRPE architecture, while preserving the original network structure.

2.5 Training Loss Functions for Synthetic and Self-Supervised Learning

Because each network is responsible for estimating different variables, the adopted loss functions must account for these differences. Thus, during synthetic training, the loss functions are defined as:

$$L_{depth} = || \mathbf{d} - \hat{\mathbf{d}} ||_1 \tag{7}$$

$$L_{joint} = || \mathbf{q} - \hat{\mathbf{q}} ||_2 \tag{8}$$

$$L_{rot} = || \mathbf{R} - \hat{\mathbf{R}} ||_2 \tag{9}$$

$$L_{trans} = || \mathbf{t} - \hat{\mathbf{t}} ||_2 \tag{10}$$

$$L_{tkps} = || \mathbf{P} - \hat{\mathbf{P}} ||_2 + || \mathbf{p} - \hat{\mathbf{p}} ||_2 \tag{11}$$

$$L'_{kps} = || \mathbf{P}' - \widehat{\mathbf{P}'} ||_2 + || \mathbf{p}' - \widehat{\mathbf{p}}' ||_2 \quad (12)$$

Where $\mathbf{d}$, $\mathbf{q}$, $\mathbf{R}$, $\mathbf{t}$, $\mathbf{P}$ and $\mathbf{P}'$ represent, respectively, the reference keypoint depth, joint position values, camera rotation matrix, camera translation, keypoint coordinates based on the robot forward kinematics, and their regressed locations. Additionally, $\mathbf{p}$ and $\mathbf{p}'$ represent the 2D projections of points $\mathbf{P}$ and $\mathbf{P}'$ onto the image plane through the camera intrinsic matrix (Ban, 2024).

From these partial loss functions, the total synthetic training loss is defined as:

$$L = L_{depth} + L_{joint} + L_{rot} + L_{trans} + L_{tkps} + L'_{kps} \quad (13)$$

For real data, training is performed using a self-supervised approach, meaning that no ground-truth labels are used. Instead, learning is driven by consistency constraints imposed by the robot's kinematic structure and the visual information present in the image, (Ban, 2024). Let $\widehat{\mathbf{P}}$ be the keypoint locations obtained from forward kinematics while respecting the robot structure, and $\widehat{\mathbf{P}'}$ be the values predicted by the model. The loss function for these variables is defined as:

$$L_{3D} = || \mathbf{P}' - \widehat{\mathbf{P}'} ||_2 \quad (14)$$

Using the values predicted by the model, the robot geometric description available in the Unified Robot Description Format (URDF) file is used to reconstruct its configuration through forward kinematics. Then, differential rendering is applied to obtain a binary segmentation mask M_{render}. Additionally, a CrNet segmentation model (Liu, 2020) is used to obtain another segmentation mask M_{seg}. The mask loss function is computed using the Intersection over Union (IoU) dissimilarity metric (Ban, 2024):

$$L_{mask} = 1 - \frac{S_{intersection}}{S_{union}} \quad (15)$$

Therefore, the total loss for this training stage is defined as:

$$L = L_{3D} + L_{mask} \quad (16)$$

3 Training

For all experiments, the model was trained using the DREAM dataset, considering only data acquired from the RealSense camera. Initially, during the synthetic training phase, DepthNet was trained using a learning rate of 1e-4 for 500 epochs, while training metrics were continuously monitored to ensure convergence and stability. After this stage, the best DepthNet weights were selected and used to initialize the complete model. The full architecture was then trained using synthetic data, allowing the evaluation of training loss and performance metrics such as AUC/ADD, which were used to determine the best model configuration.

Unlike the original model, our approach reduces the synthetic dataset to only 8000 images and corresponding features, aiming to decrease computational cost and training time while maintaining competitive performance.

Following synthetic training, the model that achieved the best AUC/ADD metric was selected to perform self-supervised training using real-world data. In this stage, the training process was performed for 400 epochs using a reduced learning rate of 1e-6, aiming to refine the model predictions and improve generalization to real scenarios. Since real data training is self-supervised, no ground-truth labels were used. Instead, the learning process relied on consistency constraints derived from the robot kinematic structure and visual consistency from the input images.

To further evaluate performance, the keypoint prediction accuracy was also analyzed using a distance-based threshold metric. For comparison purposes, a threshold value of 60 mm was adopted (Ban, 2024). When comparing the obtained results with those reported in the literature, the proposed method achieved a slightly higher accuracy than the HPRE model (Ban, 2024), demonstrating the effectiveness of incorporating depth-aware features and the adopted training strategy.

Table 1. Training Configuration Table.

Training Step	Model	Data Type	Learning Rate	Epochs
Step 1	DepthNet	Synthetic	1e-4	500
Step 2	Full Model	Synthetic	1e-4	400
Step 3	Full Model	Real	1e-6	400

4 Results

The obtained results demonstrate that the proposed method achieves competitive performance in monocular robot pose estimation. The evaluation based on keypoint prediction accuracy, using a 60 mm threshold (Ban, 2024), shows that the model provides reliable spatial localization of robot joints for practical applications.

When compared with methods reported in the literature, the proposed approach achieved an accuracy of 75.4%, slightly outperforming RoboPose (74.3%) and HPRE (75.2%). Although the improvement is modest, it is relevant given the complexity of the task and the challenges associated with monocular depth and pose estimation.

Overall, the results indicate that the proposed approach is effective and capable of achieving state-of-the-art performance while maintaining robustness in real-world scenarios.

5 Conclusion

The results demonstrate that the proposed method achieves competitive performance in monocular robot pose estimation, providing reliable spatial localization of robot joints and slightly outperforming existing approaches. The integration of depth-aware features extracted from monocular RGB images improves the model's ability to learn geometric relationships between visual data and robot pose parameters, contributing to robust performance in real-world scenarios.

A key contribution of this work is the incorporation of the Depth Anything network, which significantly reduces the dependence on large synthetic datasets. By enriching feature representation with depth information, the model is able to maintain high performance while using a reduced synthetic dataset of only 8k samples. This reduction lowers computational cost and training time, demonstrating that depth-enhanced feature extraction is an effective strategy for improving efficiency in monocular robot pose estimation tasks.

Acknowledgments. The authors thank FUNAPE – Research Support Foundation (CNPJ No. 00.799.205/0001-89) for financial support through the scholarship granted for the project "Implementation of CERISE – Center of Excellence in Wireless Intelligent Networks and Advanced Services" (cost center 08.193), carried out at the Federal University of Goiás – UFG and coordinated by Flavio Henrique Teles Vieira.

Disclosure of Interests. The authors declare no conflicts of interest relevant to the content of this article.

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